RESEARCH PAPER · KV CACHE COMPRESSION

SAGE-v2: Provably Budget-Consistent, Layer-Adaptive KV-Cache Eviction for Memory-Efficient Long-Context Inference

Correcting the budget-consistency flaw of entropy-exponent allocation with a capped water-filling projection — exact global budgets at every decoding step, decayed importance scoring with provable separation, and a matched-entry, fully traceable evaluation on multi-head and grouped-query attention models.

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Author list withheld for double-blind review · code, logs, and accounting released with the paper

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WikiText-2 · Needle Retrieval · Budget Compliance